

MUHAMMAD MAROOF

Karachi, Pakistan | github.com/maroofiums | linkedin.com/in/maroofiums | kaggle.com/maroofiums | leetcode.com/maroofiums

SUMMARY

ML/AI Engineer with hands-on experience in deep learning, NLP, and production API development. Authored independent research on learnable attention normalization in Transformer architectures (NeurIPS 2026 workshop, under review). Proficient in PyTorch, HuggingFace Transformers, LangChain, FastAPI, and MLOps tooling. Focused on bridging research-grade ML with scalable backend systems.

EDUCATION

Memon Industrial and Technical Institute (MITI)
Diploma of Associate Engineering — Software Engineering
• **Relevant Coursework:** Data Structures & Algorithms, Object-Oriented Programming, Database Systems, Computer Networks, Software Engineering Principles, Operating Systems, Discrete Mathematics.
• **Independent Study:** Deep Learning (PyTorch), Transformer Architectures, Reinforcement Learning, Generative AI, MLOps, System Design for ML, and DSA patterns for technical interviews.

Karachi, Pakistan
Expected Oct 2026

EXPERIENCE

Developer Hub Corporation
ML/AI Engineer Intern
• Designed and fine-tuned NLP pipelines using HuggingFace Transformers for text classification and information extraction.
• Built and deployed REST APIs with FastAPI, containerized via Docker, and integrated into CI/CD pipelines using GitHub Actions.
• Implemented MLflow experiment tracking for hyperparameter tuning and model versioning across training runs.

Remote
November 2025 - January 2026

Elevvo PathWays
ML API Intern / ML Specialist (Contract)
• Developed ML-powered REST APIs integrating LangChain with HuggingFace and Mistral LLM backends.
• Built a RAG (Retrieval-Augmented Generation) pipeline for document question-answering using vector stores and embedding models.
• Served fine-tuned models via FastAPI endpoints with low-latency inference optimized for production workloads.

Remote
January 2026 - March 2026

RESEARCH

Adaptive Activation Attention (A³) | Independent Research — Under Review
• Proposed replacing fixed softmax in Transformer self-attention with a learnable, input-conditioned gated transformation that adapts attention distributions per input context.
• Evaluated with bootstrap confidence intervals and explicit p-value reporting; includes null result analysis for scientific transparency. Submitted to NeurIPS 2026 workshop (Empirical Findings track).
• **Links:** Zenodo DOI | Kaggle Notebook | GitHub

May 2026

PROJECTS

Pneumonia Detection — Chest X-Ray Classification | PyTorch, OpenCV, Grad-CAM, FastAPI | GitHub
• Trained a convolutional neural network with PyTorch/torchvision to classify chest X-ray images for pneumonia detection, with an OpenCV-based image preprocessing pipeline.
• Integrated Grad-CAM for visual interpretability of model predictions and served inference through a FastAPI endpoint.

2026

Human Activity Recognition — CNN-LSTM | PyTorch, MLflow, FastAPI, Docker | GitHub
• Built a hybrid CNN-Bidirectional LSTM architecture in PyTorch to classify six human activities from UCI HAR smartphone accelerometer/gyroscope signals, achieving 90–94% test accuracy.
• Implemented subject-wise train/validation splitting to prevent data leakage, tracked experiments with MLflow, and deployed a Dockerized FastAPI serving layer.

2026

Movie Recommender — Siamese Transformer + FAISS | PyTorch, Transformers, FAISS, FastAPI | GitHub
• Designed a Siamese Transformer network trained with contrastive/metric learning to generate dense semantic movie embeddings from the TMDB Movies Metadata dataset.
• Indexed embeddings with FAISS for fast Top-K similarity search and exposed recommendations through a FastAPI backend.

2026

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript (React), Bash
ML / DL: PyTorch, TensorFlow, Scikit-learn, HuggingFace Transformers, LangChain, OpenCV
Architectures: Transformers, BERT, GPT, Encoder-Decoder, CNN, LSTM, RAG, Reinforcement Learning
Backend: FastAPI, Django REST Framework, PostgreSQL, REST API design
MLOps / DevOps: MLflow, Docker, GitHub Actions (CI/CD), AWS (S3, EC2)
Tools: Git, LaTeX, Jupyter, Linux, Kaggle, arXiv